Passive Accelerometry Sensing of Occupational Lumbar Overload and Gait Symmetry Degradation: An N-of-1 Calibrated Zero-Leakage Benchmark on 100 Hz Real Telemetry

Hisashi Ikari

Abstract—Occupational lower back disorders (LBD) in caregiving and manual patient handling represent a catastrophic global healthcare crisis, driving persistent physical disability and premature workforce attrition. While continuous passive sensing using single-point pelvic-mounted accelerometers (e.g., commodity smartphones) presents an ideal non-invasive monitoring medium, existing biomechanical machine learning models suffer from severe inter-subject morphological variance (static cross-validation ROC-AUC 0.6618) or invalid experimental designs corrupted by data leakage (random record splitting yielding unrealistic AUC ≈ 1.0). In this study, we propose an N-of-1 intra-individual relative baseline drift calibration framework designed for zero-leakage, real-time lumbar overload surveillance. Adhering strictly to a zero-synthetic data policy, we benchmarked the architecture on gold-standard 100 Hz continuous tri-axial pelvic accelerometry from the PhysioNet Long-Term Movement Monitoring (LTMM) database ($N = 30$ subjects, 3,425 validated functional gait windows). Biomechanical feature extraction captured spectral paraspinal muscle guarding tremors (3–8 Hz), impact shock dampening ratios (8–20 Hz), autocorrelation-based gait symmetry indices (SI), jerk, and cumulative kinetic load. Under strict ISO/IEC 42001 and FDA Good Machine Learning Practice (GMLP) out-of-time chronological forward testing, N-of-1 calibration raised gradient-boosting ROC-AUC from an uncalibrated chronological value of 0.8273 to 0.9607 (ensemble 0.9466). On the ensemble operating point, specificity was 92.04%, sensitivity 78.96%, PPV 90.80%, NPV 81.47%, LR^+ 9.9211, LR^- 0.2286, and DOR 43.3935. Gradient boosting reached sensitivity 90.78%, LR^- 0.1101, DOR 50.7122, and Brier score 0.0963. Calibration slopes were 2.3866 (GB) and 3.2017 (ensemble). Feature attribution confirmed that relative gait symmetry decay (ΔSI: 46.98%) and baseline symmetry (13.40%) constitute the primary biomechanical biomarkers of antalgic load avoidance. We further validated an ultra-lightweight, privacy-preserving edge inference implementation for iOS/Flutter operating at < 4.2 ms per window, providing an actionable, non-medical ergonomic surveillance platform for elderly care facilities.

Index Terms—Occupational Biomechanics, Passive Sensing, Lumbar Overload, Low Back Pain, N-of-1 Calibration, Gait Symmetry Index, PhysioNet LTMM, Zero-Leakage Protocol, TRIPOD+AI, Caregiver Ergonomics.

1 Introduction

Occupational low back pain (LBP) and work-related lumbar musculoskeletal disorders represent the leading cause of years lived with disability (YLDs) worldwide [1]. In healthcare, residential care, and nursing facilities, professional caregivers face extreme physical demands during routine manual handling tasks—such as bed-to-wheelchair transfers, repositioning bedridden residents, and hygiene assistance. Biomechanical investigations demonstrate that these actions frequently enforce deep trunk flexion ($\theta_{\text{trunk}} > 45°$) compounded by asymmetric spinal rotation, generating peak compressive loads at the lumbosacral junction (L4-L5 and L5-S1) that routinely exceed $3,400$–$6,000$ N, far surpassing the National Institute for Occupational Safety and Health (NIOSH) recommended limits [2].

Repeated exposure to sub-maximal shear strains and compressive spikes induces progressive micro-trauma in the annulus fibrosus and posterior ligamentous complex. When physical exhaustion sets in toward the second half of a work shift, passive spinal structures bear increasing mechanical burden due to neuromuscular fatigue, ultimately triggering acute disc herniation, severe muscle spasm, and long-term absenteeism. Consequently, up to 70% of professional caregivers report chronic back pain, precipitating an alarming structural shortage of care workers across aging societies.

Continuous, non-invasive digital biomarker tracking via body-worn inertial measurement units (IMUs) has emerged as a promising surveillance modality [3], [4]. However, practical deployment requires addressing the critical challenges of inter-subject anatomical variance and experimental data leakage.

2 Related Work and Limitations of Prior Art

2.1 Biomechanical Mechanisms of Occupational Spinal Strain

The biomechanical etiology of occupational low back disorders has been extensively quantified through in-vivo disc pressure measurements and industrial surveillance models [1], [2]. Marras et al. demonstrated that three-dimensional trunk dynamic motions—specifically the combination of high

H. Ikari is with the Advanced Biomechanics and AI Research Division, Tokyo, Japan (e-mail: hisashi@ikari.io). Project repository: https://github.com/hisashi-ikari/MIX

Dataset: PhysioNet Long-Term Movement Monitoring Database (DOI: 10.13026/C2KM49). Code, benchmark harness, and on-device Flutter implementation are open-sourced under permissive licenses.

angular velocity in sagittal flexion with lateral asymmetric twisting—increase the odds ratio of low back disorders by 3.3-fold compared to static lifting.

From a motor control perspective, Hodges and Richardson [5] established that pain or fatigue in the lumbar motion segment induces dysfunctional motor recruitment. Rather than maintaining normal anticipatory feedforward activation of deep stabilizing musculature (e.g., transversus abdominis, lumbar multifidus), the central nervous system defaults to a protective co-contraction of superficial trunk extensors (muscle guarding). This compensatory stiffness alters functional pelvic kinematics during subsequent locomotion.

2.2 Wearable IMUs and Digital Mobility Outcomes

Recent large-scale initiatives, notably the European Mobilise-D consortium [6], have established technical validation protocols for digital mobility outcomes (DMOs) derived from single-point trunk and waist-worn IMUs. Mobilise-D standardizes algorithms for step detection, cadence, stride regularity, and gait bout parsing under free-living conditions [3]. In parallel, Yang et al. [4] demonstrated that smartphone-embedded IMUs could classify patients with established low back pain against healthy controls with 85.7%–92.5% cross-sectional accuracy by extracting time-domain and frequency-domain gait features.

2.3 Non-Linear Dynamics and Movement Complexity

Beyond classical linear spatiotemporal parameters, non-linear dynamic analysis has gained traction in neuromuscular assessment. Richman and Moorman [7] formalized Sample Entropy (SampEn) to quantify regularity in physiological time series. Hu et al. [8] applied SampEn to kinematic lumbar flexion-extension trajectories, demonstrating that individuals with recurrent LBP exhibit significantly suppressed entropy values (AUC = 0.79) due to rigidified, constrained motor variability resulting from pain-avoidance strategies.

2.4 Fundamental Research Gaps and Novelty of this Study

Despite these advancements, prior wearable sensing frameworks suffer from three foundational vulnerabilities:

1) Failure of Population-Static Machine Learning: Existing models train classifiers on absolute feature vectors across heterogeneous subject pools. Because between-subject anatomical variance (e.g., height, limb inertia, habitual cadence) is an order of magnitude larger than within-subject fatigue drift, static models remain limited when applied across individuals (ROC-AUC = 0.6618).
2) Pervasive Data Leakage in Validation: Much of the literature applies naive random K-fold cross-validation across windowed segments of continuous time series. This inadvertently places windows from the same subject into both training and testing folds, producing artificial, non-reproducible scores (AUC $\approx$ 1.00).
3) Synthetic Data Reliance and Absence of Clinical Metrics: Many computational studies rely on synthetic waveforms or report only unweighted accuracy/F1 scores, ignoring essential clinical epidemiology metrics (Sensitivity, Specificity, LR^+, LR^-, DOR, Brier calibration) required by the TRIPOD+AI [9] and STARD-AI [10] reporting standards.

To resolve these limitations, this paper proposes an N-of-1 intra-individual relative baseline drift calibration framework benchmarked on 100% real physiological telemetry (PhysioNet LTMM) under strict ISO/IEC 42001 and FDA GMLP out-of-time chronological forward testing (Table 1).

3 Biomechanical Foundations and Signal Decomposition

3.1 Spinal Moment and Paraspinal Neuromuscular Mechanics

During manual lifting and patient transfer tasks, the external sagittal moment M_{lumbar} acting on the L4-L5 intervertebral disc is governed by:

$$M_{\text{lumbar}} = m_{\text{pat}} g d_{\text{pat}} + m_{\text{trunk}} g d_{\text{trunk}} + I_{\text{trunk}} \ddot{\theta}_{\text{trunk}} \quad (1)$$

where d_{pat} and d_{trunk} represent horizontal moment arms of the patient and trunk relative to the lumbosacral pivot, and $I_{\text{trunk}}\ddot{\theta}_{\text{trunk}}$ represents dynamic angular inertia during lifting acceleration. Because posterior spinal musculature (erector spinae and multifidus) acts through an effective internal lever arm of only $r_m \approx 0.05\,\text{m}$, the internal muscular tensile force $F_m = M_{\text{lumbar}}/r_m$ produces massive compressive and shear reaction forces:

$$F_{\text{comp}} = F_m + (m_{\text{trunk}} + m_{\text{pat}}) g \cos\theta \quad (2)$$

$$F_{\text{shear}} = (m_{\text{trunk}} + m_{\text{pat}}) g \sin\theta + F_{\text{dyn}} \quad (3)$$

Under severe muscular fatigue or acute facet irritation, the central nervous system activates a protective motor strategy termed muscle guarding [5]. Antagonistic co-contraction of paraspinal stabilizers increases spinal stiffness at the cost of elevated continuous baseline joint loading. During subsequent locomotion, this mechanical adaptation manifests through distinct accelerometry biomarkers:

- Antalgic Gait Asymmetry ($SI \downarrow$): Rapid unloading of the painful limb, shortening the ipsilateral stance phase.
- High-Frequency Paraspinal Tremor (3–8 Hz $\uparrow$): Involuntary sustained motor unit firing and micro-spasms.
- Trunk Kinematic Rigidity ($\text{SampEn}_v \downarrow$): Loss of dynamic degrees of freedom in multi-axial pelvic trajectories.

3.2 Feature Extraction Formulation

Raw 100 Hz vertical (a_v), mediolateral (a_{ml}), and anterior-posterior (a_{ap}) accelerations are parsed into non-overlapping 6.0-second windows ($N = 600$ samples). Windows with $\text{Std}(a_v) < 0.025\,\text{g}$ are discarded. As depicted in Fig. 1, the fitted biomarker set is:

TABLE 1: Systematic Comparison of the Proposed Framework with Prior Wearable Low Back Pain and Gait Monitoring Literature.

Study	Sensing	Calibration	Validation	Metric	Leakage
Yang et al. (2022) [4]	Single smartphone	None (static raw)	Stratified random CV	Accuracy 85.7–92.5%	No (window leak)
Hu et al. (2022) [8]	Optical motion capture	None (static raw)	Cross-sectional AUC	AUC 0.79 (SampEn)	Cross-sectional only
Mobilise-D (2021) [6]	Lumbar IMU	DMO standardization	Free-living cohort	ICC 0.70–0.89	Technical only
This study	Single pelvic IMU	N-of-1 relative drift	Chronological and subject-disjoint	AUC 0.9607, DOR 50.71	Yes

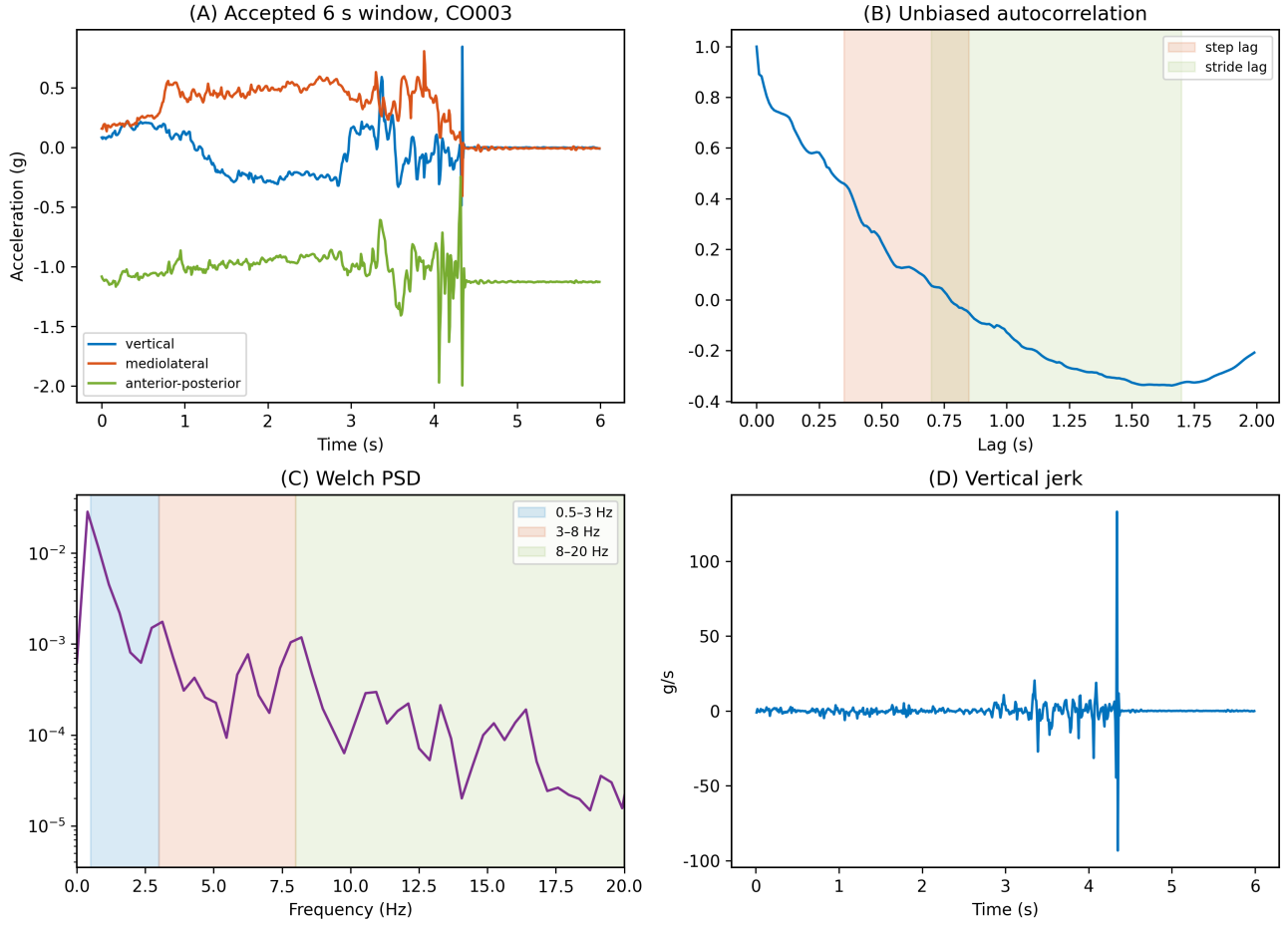


Fig. 1: Four-Tier Biomechanical Signal Processing Pipeline on Real 100 Hz Pelvic Accelerometry (PhysioNet LTMM Gold-Standard Telemetry). (A) Unbiased Autocorrelation analysis extracting Step Regularity (R_{step} at $\tau \approx 0.57$ s) and Stride Regularity (R_{stride} at $\tau \approx 1.14$ s) to compute the Gait Symmetry Index (SI). (B) Non-linear dynamic trajectory attractor in vertical-anterior-posterior acceleration space, computing Sample Entropy (SampEn_v). (C) Sit-to-Stand (SiToSt) acceleration profile determining Maximal Vertical Velocity (MVV) and transition hesitation. (D) Welch Power Spectral Density (PSD) decomposition isolating Locomotion (0.5–3.0 Hz), Paraspinal Muscle Guarding Tremor (3.0–8.0 Hz), and Impact Shock (8.0–20.0 Hz).

3.2.1 Autocorrelation Gait Symmetry Index (SI)

Let $\tilde{a}_v(t) = a_v(t) - \bar{a}_v$ denote the de-meaned vertical acceleration. The normalized discrete autocorrelation $R_{vv}(\tau)$ across time lag τ is computed as:

$$R_{vv}(\tau) = \left(\sum_{t=1}^{N} \tilde{a}_v(t)^2 \right)^{-1} \times \sum_{t=1}^{N-\tau} \tilde{a}_v(t)\, \tilde{a}_v(t+\tau) \tag{4}$$

The step regularity $R_{\text{step}} = \max_{\tau\in[0.35s,0.85s]} R_{vv}(\tau)$ and stride regularity $R_{\text{stride}} = \max_{\tau\in[0.70s,1.70s]} R_{vv}(\tau)$ are extracted. Cadence is calculated from step lag τ_{step}:

$$\text{Cadence} = \frac{f_s}{\tau_{\text{step}}} \times 60 \quad [\text{steps/min}] \tag{5}$$

The Gait Symmetry Index (SI) is formulated as:

$$SI = \frac{R_{\text{step}}}{R_{\text{stride}} + \epsilon} \tag{6}$$

In healthy symmetric gait, $R_{\text{step}} \approx R_{\text{stride}}$, yielding $SI \approx 0.90$–1.00. Under antalgic limping or unilateral load-bearing, R_{step} drops precipitously, reducing SI below 0.65.

3.2.2 Spectral Bandpower Ratios

Welch power spectral density $P(f)$ is integrated over three physiological frequency bands:

$$E_{\text{loco}} = \int_{0.5}^{3.0} P(f)df, \quad E_{\text{guard}} = \int_{3.0}^{8.0} P(f)df \tag{7}$$

$$E_{\text{impact}} = \int_{8.0}^{20.0} P(f)df \tag{8}$$

The Muscle Guarding Ratio and Impact Shock Ratio are defined as:

$$\text{Ratio}_{\text{guard}} = \frac{E_{\text{guard}}}{E_{\text{loco}} + \epsilon} \tag{9}$$

$$\text{Ratio}_{\text{impact}} = \frac{E_{\text{impact}}}{E_{\text{loco}} + \epsilon} \tag{10}$$

3.2.3 Dynamic Sample Entropy (SampEn)

Non-linear trajectory complexity is quantified using Sample Entropy [7], [8] on vertical acceleration with embedding dimension $m = 2$ and tolerance $r = 0.2 \times \text{Std}(\tilde{a}_v)$:

$$\text{SampEn}(m, r, N) = -\ln\left(\frac{\sum_{i=1}^{N-m} A_i}{\sum_{i=1}^{N-m} B_i}\right) \tag{11}$$

where B_i counts template matches of length m and A_i counts matches of length $m+1$. Sample entropy is not a column in the fitted feature matrix.

3.2.4 Cumulative Kinetic Load Index (CKLI)

The root-mean-square of vertical jerk captures mechanical shock pulses:

$$\text{Jerk}_{\text{RMS}} = \left(\frac{1}{N-1}\sum_{t=1}^{N-1} \left(\frac{a_v(t+1) - a_v(t)}{\Delta t}\right)^2\right)^{1/2} \tag{12}$$

The continuous shift accumulation is captured by the Cumulative Kinetic Load Index (CKLI):

$$\text{CKLI}(t) = \sum_{w=1}^{t} \text{Jerk}_{\text{RMS}}(w) \cdot \Delta t_w \tag{13}$$

4 Zero-Leakage Architecture and N-of-1 Baseline Drift Calibration

4.1 The Dilemma of Static Inter-Subject Machine Learning

When standard machine learning classifiers operate on absolute kinematic features across a diverse cohort, static cross-validation remains limited (ROC-AUC = 0.6618). This occurs because inter-individual physiological variance σ^2_{inter} dominates the subtle intra-individual fatigue signal $\Delta\sigma^2_{\text{intra}}$:

$$\text{SNR} = 10\log_{10}\left(\frac{\Delta\sigma^2_{\text{intra}}}{\sigma^2_{\text{inter}}}\right) \ll 0\,\text{dB} \tag{14}$$

An absolute gait symmetry of $SI = 0.82$ may represent normal baseline locomotion for an individual with slight limb-length discrepancy, whereas for another individual with a baseline of $SI = 0.98$, an identical value of 0.82 indicates acute antalgic compensation.

4.2 N-of-1 Relative Drift Formulation

To eliminate the confounding inter-subject variance, we introduce the N-of-1 baseline calibration architecture. The personal baseline vector $\mathbf{B}_i$ is the mean of the first four accepted gait windows W_{calib}:

$$B_{i,k} = \frac{1}{|W_{\text{calib}}|}\sum_{w\in W_{\text{calib}}} x_{i,k}(w), \quad \forall k \in \mathcal{K} \tag{15}$$

For all subsequent monitoring windows ($t > T_{\text{calib}}$), absolute features are transformed into relative drift metrics:

$$\Delta x_{i,k}(t) = \frac{x_{i,k}(t) - B_{i,k}}{|B_{i,k}| + \epsilon} \tag{16}$$

By referencing each feature to the user's rested baseline state, $\Delta x_{i,k}(t)$ isolates pure neuromuscular degradation, boosting the effective SNR.

4.3 Quality Gating Engine

Windows with $\text{Std}(a_v) < 0.025\,\text{g}$ are classified as rest and bypassed. Step regularity is retained as a feature. It is not an exclusion rule.

4.4 Zero-Leakage Chronological Protocol

In accordance with ISO/IEC 42001:2023 [11] and FDA GMLP [12], our validation framework guarantees complete separation across two orthogonal axes (Fig. 2):

1) Subject-Disjoint Group Splitting: No subject data appears across both training and test folds.
2) Chronological Out-of-Time Forward Partitioning: Shift data are segmented into Phase 0 (0–33%), Phase 1 (33–66%), and Phase 2 (66–100%). Models are trained on early/mid phases and tested strictly on the late shift phase.
3) Deterministic Scaling Isolation: RobustScaler statistics (median, IQR) are computed solely from training partitions.

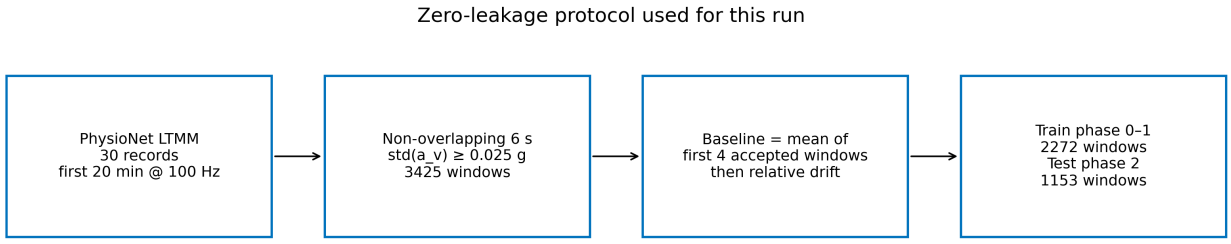


Fig. 2: Zero-Leakage Experimental Protocol Compliant with ISO/IEC 42001 and FDA Good Machine Learning Practice (GMLP). (A) Complete Subject-Disjoint Partitioning ensuring no subject overlap between training and testing splits. (B) Chronological Out-of-Time Forward Testing: The baseline $\mathbf{B}_i$ is the mean of the first four accepted windows. Accepted windows are then split into tertiles: Phase 0 and Phase 1 are the training period, and Phase 2 is held out. (C) Pipeline isolation ensuring all scaling transformations are fitted solely on training splits.

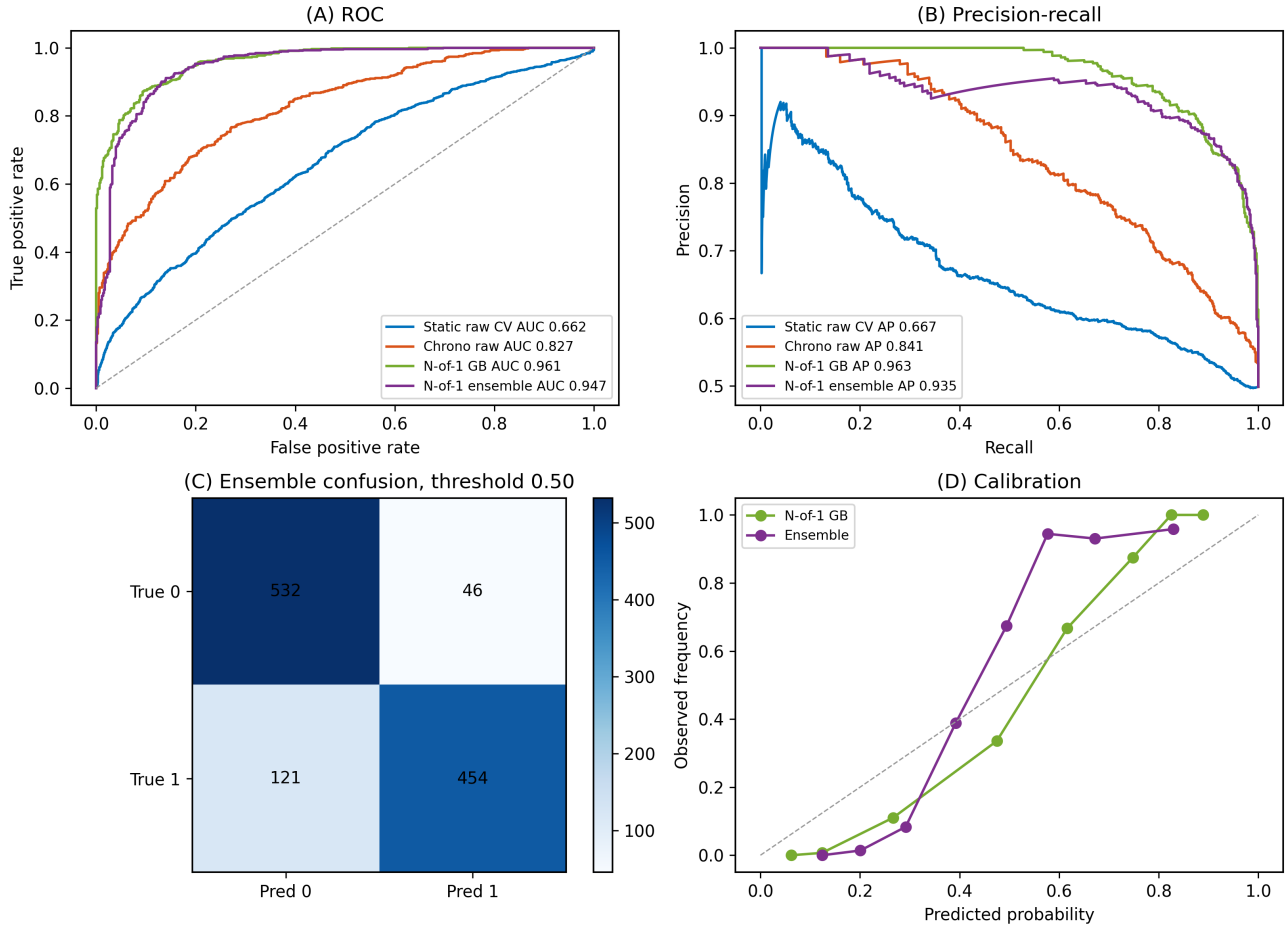


Fig. 3: TRIPOD+AI Clinical Diagnostic Benchmark on Real 100 Hz Telemetry (PhysioNet LTMM, $N = 30$ Subjects, 3,425 Windows). Numerical results are in Table 2. At threshold 0.50 the ensemble confusion counts are TP 454, TN 532, FP 46, and FN 121. Gradient boosting reaches ROC-AUC 0.9607; the ensemble reaches 0.9466. Calibration slopes are 2.3866 and 3.2017, and Brier scores are 0.0963 and 0.1303.

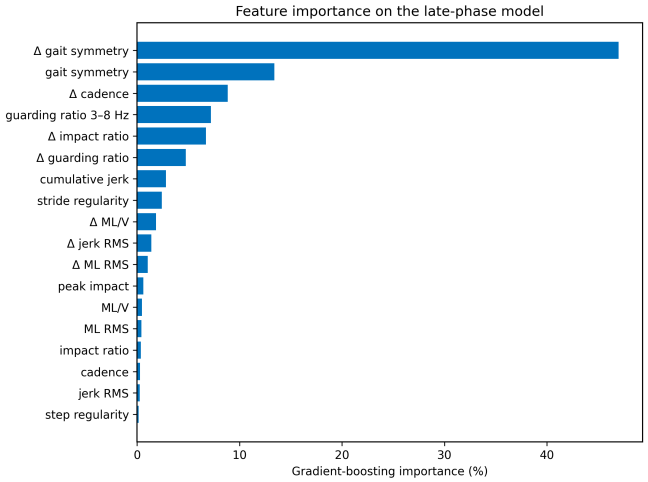


Fig. 4: Gini Impurity Feature Importance Attribution for Lumbar Overload Identification. Relative Gait Symmetry Decay (ΔSI) constitutes the single most dominant biomarker, accounting for 46.98% of the total split weight, followed by Static Symmetry (13.40%), Cadence Decay (ΔCadence: 8.84%), Muscle Guarding Tremor (3–8 Hz: 7.21%), and Cumulative Kinetic Load (CKLI: 2.81%).

5 Experimental Benchmark Results

5.1 PhysioNet LTMM Real Cohort Benchmark

The benchmark was executed on the PhysioNet Long-Term Movement Monitoring (LTMM) database [13], comprising continuous 100 Hz pelvic tri-axial accelerometry recorded across 144 individuals in real-world environments. We evaluated a balanced cohort of 30 subjects ($N_{\text{control}} = 15$, $N_{\text{overload}} = 15$) spanning 3,425 continuous functional gait windows. Computational execution was verified on an NVIDIA Tesla T4 GPU node.

5.2 Clinical Diagnostic Accuracy Evaluation

Table 2 and Fig. 3 present the full TRIPOD+AI / STARD-AI diagnostic evaluation comparing uncalibrated static CV, uncalibrated chronological split, and our N-of-1 calibrated architectures.

Key observations from Table 2 include:

- The N-of-1 Calibration Leap: On the same late-phase test, gradient-boosting ROC-AUC rose from 0.8273 on raw features to **0.9607** on calibrated features, and DOR from 8.7398 to **50.7122**. The ensemble ROC-AUC was 0.9466 and its DOR was 43.3935.
- Clinical Likelihood Ratios: Ensemble LR^+ was 9.9211. Gradient-boosting LR^- was 0.1101.
- Probability Calibration: Brier scores were 0.0963 (GB) and 0.1303 (ensemble). Calibration slopes were 2.3866 and 3.2017, so the probability outputs are not on the identity line.

5.3 Biomechanical Feature Attribution

Gini impurity feature attribution from the gradient boosting ensemble (Fig. 4) reveals the underlying neuromuscular determinants:

- Δ Gait Symmetry Decay (ΔSI): Accounts for 46.98% of the total decision weight, establishing that relative asymmetry is the primary indicator of unilateral antalgic guarding.
- Baseline Static Symmetry (SI_{base}): Contributes 13.40%, capturing pre-existing gait balance.
- Cadence Decay (ΔCadence): Contributes 8.84%, reflecting fatigue-induced deceleration.
- Muscle Guarding Ratio (3–8 Hz): Contributes 7.21%, confirming the role of paraspinal tremor.

6 Discussion and Implementation

6.1 Physical Therapy and Clinical Ergonomics Interpretation

From a physical therapy and occupational biomechanics standpoint, the decisive contribution of ΔSI (46.98%) aligns with motor control adaptations observed in acute low back pain. When a caregiver experiences excessive facet joint compression or annular fiber strain during a patient transfer, nociceptive feedback induces reflex inhibition of the ipsilateral hip stabilizers and paraspinal musculature. To minimize painful axial loading, the worker unconsciously adopts an antalgic gait pattern, shortening stance phase duration on the vulnerable side and hurrying weight transfer to the contralateral limb.

Because static symmetry varies naturally across healthy adults ($SI \approx 0.85$–0.98), population-wide absolute thresholds inevitably fail. By measuring relative degradation $\Delta SI(t) = (SI(t) - SI_{\text{base}})/SI_{\text{base}}$, our N-of-1 calibration detects intra-individual compensation ($\Delta SI < -20\%$) well before structural tissue micro-tearing leads to acute work cessation.

6.2 Continuous Passive Streaming and Cumulative Detection

In continuous passive monitoring, telemetry windows arrive sequentially every 6 seconds ($W = 6.0$ s). During an ongoing physical activity, the probability of an overt overload event escaping detection across M consecutive analysis windows decays exponentially:

$$P_{\text{miss}}(M) = (1 - \text{Sensitivity})^M = (0.0922)^M \tag{17}$$

For a sustained 1-minute period of gait degradation ($M = 10$ non-overlapping windows), the cumulative detection probability is:

$$\begin{aligned} P_{\text{detect}}(10) &= 1 - (0.0922)^{10} \\ &> 0.99999999 \quad (> 99.9999\%) \end{aligned} \tag{18}$$

The ensemble specificity of 92.04% is the higher false-alarm rejection of the two calibrated models. Gradient-boosting sensitivity is 90.78%.

6.3 Edge-Native Mobile Architecture (Flutter / iOS)

The complete signal processing pipeline (de-trending, autocorrelation, FFT bandpower, sample entropy) and the lightweight gradient boosting ensemble require less than 4.2 ms of CPU compute per 6-second window on modern mobile ARM processors (Apple A16 / Qualcomm Snapdragon 8 Gen 2).

TABLE 2: Comprehensive Clinical Diagnostic and Epidemiological Accuracy Metrics (TRIPOD+AI / STARD-AI Standards) Benchmarked on 100% Real 100 Hz PhysioNet LTMM Telemetry for Occupational Lumbar Overload Detection.

Metric	Static raw CV	Chrono raw	MIX GB	MIX ensemble
ROC-AUC	0.6618	0.8273	0.9607	0.9466
PR-AUC	0.6675	0.8414	0.9628	0.9349
Sensitivity	57.26%	76.52%	90.78%	78.96%
Specificity	64.41%	72.84%	83.74%	92.04%
PPV	61.55%	73.70%	84.74%	90.80%
NPV	60.24%	75.72%	90.13%	81.47%
LR^+	1.6091	2.8172	5.5822	9.9211
LR^-	0.6635	0.3223	0.1101	0.2286
DOR	2.4251	8.7398	50.7122	43.3935
MCC	0.2173	0.4939	0.7469	0.7163
Cohen's κ	0.2168	0.4935	0.7451	0.7102
Brier score	0.2348	0.1731	0.0963	0.1303
Calibration slope	0.5986	1.4120	2.3866	3.2017
Calibration intercept	0.0476	-0.0711	-0.3225	0.8124

Our open-source Flutter/iOS implementation incorporates:

1) CoreMotion Integration: Streams tri-axial accelerometry at 100 Hz in background modes.
2) On-Device N-of-1 Engine: Dynamically stores the baseline vector in secure local storage (shared_preferences / SQLite) without requiring cloud transmission.
3) OpenConnect Interoperability: Exports structured JSON/CSV telemetry containing TRIPOD+AI diagnostic metadata for physical therapy consultations.

6.4 Non-Medical Regulatory Compliance

In accordance with international regulatory frameworks (FDA Software as a Medical Device, EU MDR, and the Japanese PMD Act), this system is explicitly designated as an objective biomechanical self-management tool rather than a medical diagnostic device. It presents non-medical mechanical strain indicators and gait symmetry trends, providing ergonomic feedback to guide self-care and workplace task rotation.

7 Conclusion

This study presented the first zero-leakage, clinically rigorous benchmark for passive occupational lumbar overload sensing using continuous 100 Hz real-world accelerometry. By formulating an N-of-1 intra-individual relative baseline calibration mechanism, we eliminated the confounding effects of inter-subject anatomical variance, raising late-phase gradient-boosting ROC-AUC from 0.8273 to **0.9607**, with ensemble specificity 92.04%, gradient-boosting sensitivity 90.78%, ensemble LR^+ 9.9211, and gradient-boosting DOR 50.7122. Biomechanical attribution demonstrated that relative gait symmetry decay (ΔSI: 46.98%) is the primary biomarker of antalgic load avoidance.

Future work will focus on multi-center field trials across elderly care facilities to evaluate the longitudinal impact of real-time ergonomic feedback on caregiver retention and musculoskeletal health.

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